# Jinil Kim

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## RESEARCH INTERESTS

- *Computational Neuroscience* (principles linking brain and behavior through shared latent structure)
- *NeuroAI and AI for Neuroscience* (brain foundation models; neuro-inspired AI)
- *Computational Psychiatry* (model-derived markers; predicting intervention response)

## EDUCATION

**Seoul National University** — Seoul, South Korea
B.S. Computer Science · B.A. Psychology · B.A. Linguistics — Mar 2021 – Exp. Feb 2027
GPA: 4.18 / 4.3 (3.97 / 4.0) · 9 semesters (Mandatory military service, Jul 2023 – Jan 2025)

**The University of Hong Kong** — Hong Kong SAR
Summer Research Program, Department of Psychology — Jun 2025 – Aug 2025

## PUBLICATIONS & MANUSCRIPTS

*Workshop Papers*

- **Kim, J.**, & Ryu, J. (2026). *Conversational relevance in the conjunction fallacy is represented independently in language models.* Accepted for Conference on Neural Information Processing Systems (NeurIPS) 2026 Workshop on Linguistic Principles for Foundation Models, Sydney, Australia. [paper]
- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. (2026). *Inferring individualized color-vision distortions from fMRI hue-representation geometry.* Presented at International Conference on Machine Learning (ICML) 2026 Workshop on Structured Data for Health, Seoul, Korea. [paper]

*Under Review*

- **Kim, J.**, Jeong, J., Yun, H., Oh, M.-H., & Ahn, W.-Y. (2026). *Inverse reinforcement learning with state-dependent foresight for human decision making.* Under review at the International Conference on Learning Representations (ICLR).

*In Preparation*

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. *Preserved categorical identification with impaired hue interpolation in color vision deficiency motivates a cortically derived correction filter.*
- **Kim, J.**, & Yao, Y.-W. *Depression is associated with weaker evidence accumulation and greater caution during social decision-making.*

## RESEARCH EXPERIENCES

**Connectome Lab** — Seoul National University
Undergraduate RA (Advisor: Prof. Jiook Cha) — Apr 2025 – Present

- Led a project characterizing distortions of cortical hue geometry in color vision deficiency from fMRI. Translated them into personalized filters that moved hue-discrimination thresholds to the control range.
- Leading uncertainty evaluation of an EEG neural-field foundation model, showing that distributional scores catch distortions that point metrics miss and that seed variance can flag unreliable channels.
- Leading tests of literature priors for brain foundation models via text alignment or positional encoding.

**Computational Clinical Science Lab** — Seoul National University
Undergraduate RA (Advisor: Prof. Woo-Young Ahn) — Apr 2025 – Present

- Led the development of SC-AIRL, an inverse RL model with state-dependent foresight, raising action accuracy by up to 10 points over AIRL. Inferred planning depth correlated with anxiety and impulsivity.
- Testing whether fMRI, EEG, HRV, and survey data jointly predict alcohol-use severity ($n = 60$) and yield interpretable markers, using multiblock sparse PLS, the regression form of DIABLO.

**MIND Lab** — University of Hong Kong
Visiting Researcher (Advisor: Prof. Yuan-Wei Yao) — Jun 2025 – Aug 2025

- Fit stationary and time-varying drift-diffusion models to social decisions ($N = 351$), linking depression to weaker evidence accumulation ($d = 0.30$), greater caution ($d = 0.40$), and later reward-emotion integration.

## CONFERENCE PRESENTATIONS

- **Kim, J.**, Cho, M., Seo, J., & Cha, J. (2026). *fMRI decoding reveals intact neural color representations in color vision deficiency.* Organization for Human Brain Mapping (OHBM) 2026 Annual Meeting, Bordeaux, France. [Poster]
- **Kim, J.**, & Cha, J. (2026). *A framework decomposing individual arousal predictions into visual-concept and neural contributions.* Submitted to Computational and Systems Neuroscience (Cosyne) 2027.

## SCHOLARSHIPS & HONORS

**KOSAF Humanity 100 Years Undergraduate Scholarship** ($12,500) Mar 2025 – Jul 2026

- National merit scholarship from the Korean Ministry of Education; full tuition and living stipend.

**Best Poster Award, SNU Undergraduate Research Program** Jan 2026
**SNU Undergraduate Research Grant** ($3,500) Apr 2025
**SNU Student-Proposed Liberal Arts Course Contest** (1st Place) Jul 2023

- Proposed course: *Interdisciplinary Explorations with a Chatbot.*

**SNU Academic Excellence Scholarship** (3 semesters) Mar 2022 – Jun 2023

## RELEVANT COURSEWORK

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| **Neuroscience** | Neuroscience · Data Science in Human Neuroimaging · Clinical Neuropsychology · Cognitive Processes · Computational Modeling in Experimental Psychology · Brain, Mind, and Behavior |
| **Computer Science** | Data Structures · Algorithms · Computer Architecture · Logic Design · Computer Programming · Computational Linguistics |
| **Mathematics & Statistics** | Linear Algebra · Discrete Mathematics · Calculus I–II · Psychological Statistics · Advanced Psychological Statistics |

## ACADEMIC SERVICES

**Conference Reviewer:** NeurIPS 2026 Workshop on Foundation Models for the Brain and Body
**Volunteer:** ICML 2026

## TEACHING EXPERIENCE

**Basic Computing: First Adventure in Computing** Seoul National University
Teaching Assistant (Instructors: Yamada Akihiko, Heajung Min) Autumn 2025 · Spring 2026
Instructed laboratory classes regarding Python and basic programming for local and international students; English-medium section in Spring 2026.

## LANGUAGES

**Programming:** Python (advanced), R, C++, Java (proficient), Stan (intermediate)